PRACTICAL 3

Diabetes Classification using K-Nearest Neighbors (KNN)

Top 60 Viva Questions & Answers | Important concepts, tricky questions, formulas and last-minute recall

Target labels: *0 = Non-diabetic | 1 = Diabetic (positive class)*

Code syntax is intentionally minimized. Focus on concepts, reasoning, calculations and interpretation.

A. Diabetes Dataset and Classification

1. What is the objective of this practical?

Answer: Classify patients as diabetic or non-diabetic using KNN, then evaluate the model using a confusion matrix, accuracy, error rate, precision and recall.

2. What type of machine learning problem is this?

Answer: Supervised binary classification: the model learns from labelled patient records and predicts one of two classes.

3. What do Outcome values 0 and 1 represent?

Answer: 0 means non-diabetic; 1 means diabetic. Class 1 is treated as the positive class in the metrics.

4. What are features and target?

Answer: Features (X) are measurements such as glucose, BMI, blood pressure and age. The target (y) is the Outcome column.

5. Why is Outcome excluded from X?

Answer: It is the answer the model must predict. Including it as an input would reveal the answer and cause target leakage.

6. Name important features in the Pima diabetes dataset.

Answer: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction and Age. Outcome is the target.

7. Why is this binary classification?

Answer: Because the output has exactly two classes: 0 (non-diabetic) and 1 (diabetic).

8. Classification versus regression?

Answer: Classification predicts categories; regression predicts continuous numerical values.

B. Data Preprocessing and Missing Values

9. Why is preprocessing important?

Answer: It prepares data for reliable model use by handling invalid values, missing data and different feature scales.

10. Why are some zeros replaced with NaN?

Answer: In this dataset, zero in measurements such as glucose, blood pressure, skin thickness, insulin and BMI can represent missing or unrecorded measurements.

11. Why are not all zeros replaced?

Answer: Some zeros are valid, such as Pregnancies = 0 and Outcome = 0. Replacing all zeros would corrupt valid information.

12. What does NaN mean?

Answer: NaN means 'Not a Number' and is commonly used to represent a missing or invalid numerical value.

13. What is imputation?

Answer: Replacing missing values with selected or estimated values so records can be used by a model.

14. What is median imputation?

Answer: It replaces missing values in a feature with that feature's median, the middle value after sorting.

15. Why use median instead of mean?

Answer: Median is less affected by extreme values or outliers and can better represent the centre of skewed measurements.

16. What is the difference between mean and median?

Answer: Mean is the sum divided by the count. Median is the middle sorted value, or the average of the two middle values when the count is even.

17. What is data leakage in preprocessing?

Answer: It occurs when test-set information influences training, making evaluation too optimistic. Fitting an imputer on the full dataset leaks test statistics.

18. What is the correct order for imputation and splitting?

Answer: Split first; fit the imputer on training data only; transform both training and test data using that fitted imputer. Apply the same principle to scaling.

C. Train-Test Split and Feature Scaling

19. Why split into training and testing sets?

Answer: Training data fits the model. Separate unseen test data estimates how well it performs on new patients.

20. What does an 80:20 split mean?

Answer: 80% of records are used for training and 20% for testing.

21. What is stratified splitting?

Answer: It maintains approximately the same proportion of each target class in training and testing sets.

22. Why is feature scaling important for KNN?

Answer: KNN uses distances. A feature with a much larger numerical range could dominate distance calculations unless features are put on comparable scales.

23. What is StandardScaler?

Answer: It standardizes each feature by subtracting its mean and dividing by its standard deviation, using statistics learned from training data.

24. Write the StandardScaler formula and define its terms.

Answer: $z = (x - \mu) / \sigma$. x is the original value, μ is the training-feature mean, σ is its training-feature standard deviation, and z is the standardized value.

25. Why fit the scaler only on training data?

Answer: To avoid using test-set information. The training mean and standard deviation are reused to transform the test data.

26. Does StandardScaler make every value lie between 0 and 1?

Answer: No. It centres values around zero and scales them to unit variance; it does not impose a fixed 0-to-1 range.

27. What is the difference between fit_transform() and transform()?

Answer: fit_transform() learns the transformation from training data and applies it. transform() applies the already learned transformation to new data, such as the test set.

D. K-Nearest Neighbors (KNN)

28. What is KNN?

Answer: A supervised algorithm that predicts a sample's class using the labels of its K closest training samples.

29. How does KNN classify a new patient?

Answer: It calculates distances from the new patient's feature vector to training patients, selects the K nearest and predicts the majority class.

30. What does K=5 mean?

Answer: The model considers the five nearest training patients when predicting the test patient's diabetes class.

31. How does KNN decide the final class?

Answer: By majority voting among the K nearest neighbours; the class with the most votes is predicted.

32. What is Euclidean distance?

Answer: A measure of straight-line distance between two points in feature space.

33. Write the Euclidean distance formula.

Answer: d(x,y) = √[Σ(i=1 to n) (x■ – y■)²], where x■ and y■ are corresponding feature values and n is the number of features.

34. Does KNN need a graph to calculate distance?

Answer: No. It calculates distances mathematically between numerical feature vectors. Points can be imagined in multidimensional feature space; the practical does not plot them.

35. What happens if K is too small?

Answer: The model can become sensitive to noise and individual examples, increasing overfitting risk.

36. What happens if K is too large?

Answer: Distant samples influence the decision and local patterns may be smoothed away, increasing underfitting risk.

37. Why is KNN called a lazy learner?

Answer: It does little model construction during training and mainly stores examples. Much computation happens when a prediction is requested.

38. What are KNN's advantages and limitations?

Answer: It is simple and intuitive, but prediction can be slow on large datasets, it is sensitive to scaling and feature relevance, and results depend on K and the distance metric.

E. Confusion Matrix and Performance Metrics

39. What is a confusion matrix?

Answer: A table comparing actual and predicted classes. It counts true positives, true negatives, false positives and false negatives.

40. What is a True Positive (TP) here?

Answer: An actually diabetic patient correctly predicted as diabetic.

41. What is a True Negative (TN) here?

Answer: An actually non-diabetic patient correctly predicted as non-diabetic.

42. What is a False Positive (FP) here?

Answer: A non-diabetic patient incorrectly predicted as diabetic.

43. What is a False Negative (FN) here?

Answer: A diabetic patient incorrectly predicted as non-diabetic.

44. What is accuracy?

Answer: The proportion of all predictions that are correct: (TP + TN) / (TP + TN + FP + FN).

45. What is error rate?

Answer: The proportion of incorrect predictions: (FP + FN) / (TP + TN + FP + FN), or 1 – accuracy.

46. What is precision?

Answer: Of all patients predicted as diabetic, the proportion actually diabetic: TP / (TP + FP).

47. What is recall?

Answer: Of all actually diabetic patients, the proportion correctly identified: TP / (TP + FN).

48. What is the difference between precision and recall?

Answer: Precision focuses on how trustworthy positive predictions are; recall focuses on how many actual positive cases are detected.

49. Why is recall important in diabetes prediction?

Answer: Low recall means more diabetic patients are missed (false negatives), which could delay further medical assessment. The model is not a diagnosis.

50. What does zero_division=0 do?

Answer: If a metric has a zero denominator, it returns 0 instead of raising a division warning.

F. Important Tricky and Applied Concepts

51. Can accuracy alone prove the model is good?

Answer: No. With imbalanced classes, a model can get high accuracy by predicting mostly the majority class. Precision, recall and the confusion matrix are also important.

52. What is an imbalanced dataset?

Answer: A dataset where one class has many more records than the other, such as many more non-diabetic than diabetic patients.

53. Which error can be especially concerning in diabetes screening?

Answer: A false negative: an actually diabetic patient is predicted non-diabetic. This can delay follow-up, so recall matters.

54. What is overfitting?

Answer: When a model learns training data and noise too closely, performing well on training examples but poorly on unseen data.

55. What is underfitting?

Answer: When a model is too simple or has not learned enough patterns, resulting in poor performance on both training and test data.

56. How can KNN performance be improved?

Answer: Tune K, compare distance metrics, select relevant features, scale features appropriately and use cross-validation.

57. What is cross-validation?

Answer: A method that divides training data into folds, repeatedly trains on some folds and validates on another to estimate performance and tune settings.

58. Why might KNN be slow during prediction?

Answer: It may calculate distances from a new sample to many stored training samples, especially with many rows and features.

59. What is the limitation of this practical's imputation step?

Answer: The imputer is fitted before the train-test split, so test-set information affects the medians. Better: split first and fit imputation only on training data.

60. Does this model diagnose diabetes?

Answer: No. It produces a machine-learning classification based on dataset patterns. Medical diagnosis requires clinical assessment and appropriate testing.

Formula Sheet – Must Remember

Core formulas for KNN distance, feature scaling and evaluation, plus useful additional metrics.

Concept	Formula
Euclidean distance	$d(x,y) = \sqrt{[\Sigma(i=1 \text{ to } n) (x■ – y■)^2]}$
StandardScaler / Z-score	$z = (x – \mu) / \sigma$
Accuracy	(TP + TN) / (TP + TN + FP + FN)
Error rate	(FP + FN) / (TP + TN + FP + FN) = 1 – Accuracy
Precision	TP / (TP + FP)
Recall / Sensitivity	TP / (TP + FN)
Specificity	TN / (TN + FP)
F1-score (additional useful metric)	2 × (Precision × Recall) / (Precision + Recall)
Mean	$\mu = (\Sigma x■) / n$

Symbols and meanings

Symbol	Meaning
TP	True Positive: diabetic patient correctly predicted diabetic.
TN	True Negative: non-diabetic patient correctly predicted non-diabetic.
FP	False Positive: non-diabetic patient incorrectly predicted diabetic.
FN	False Negative: diabetic patient incorrectly predicted non-diabetic.
x	Original feature value.
μ	Mean of the feature, calculated from training data.
σ	Standard deviation of the feature, calculated from training data.
n	Number of features (distance formula) or values (mean formula).
K	Number of nearest neighbours used by KNN.

Last-Minute Recall

- **Target:** Outcome 0 = non-diabetic; Outcome 1 = diabetic (positive class).
- **Preprocessing:** Zero in selected medical measurements can indicate missing data; do not replace valid zeros such as Pregnancies = 0 or Outcome = 0.
- **Median imputation:** Replaces missing values with the feature median and is less affected by outliers than the mean.
- **Leakage-safe order:** Split first; fit imputer and scaler on training data only; transform test data using the fitted objects.
- **Scaling:** Important for distance-based KNN because features can have different numeric ranges.
- **KNN:** Calculate distances, select K nearest training samples, use majority voting. This practical uses K = 5.

• **Confusion matrix:** TP = diabetic correctly found; TN = non-diabetic correctly found; FP = non-diabetic wrongly called diabetic; FN = diabetic missed.

• **Metrics:** Accuracy = overall correctness; error rate = incorrect fraction; precision = reliability of positive predictions; recall = fraction of actual positives detected.

• **Medical context:** False negatives can delay follow-up, so recall matters. The model is not a clinical diagnosis.

• **Graph:** No graph is plotted in the code. KNN computes distances internally in a multidimensional feature space.